

Chapter 4 - Fouriers Method II

4.1

Solve the heat conduction problem

$$\begin{aligned}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, \quad t > 0, x \in \mathbb{R} \\ u(x, 0) &= e^{-x^2}, \quad x \in \mathbb{R}\end{aligned}$$

Solution:

The solution will take the form

$$u(x, t) = \int_{\mathbb{R}} T(\xi, t) e^{ix\xi} d\xi$$

We have the relation

$$2\pi T(\xi, t) = (\mathcal{F}_x u)(\xi, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ix\xi} dx$$

where $\mathcal{F}_x u$ is the partial Fourier transform of u in x . $u(x, 0) = e^{-x^2}$ is bounded. Taking the partial Fourier transform of the differential equation yields

$$\int_{-\infty}^{\infty} \frac{\partial u}{\partial t} e^{-ix\xi} dx - \int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial x^2} e^{-ix\xi} dx = 0 \quad (1)$$

We now assume it's possible to swap integration and derivation

$$\int_{-\infty}^{\infty} \frac{\partial u}{\partial t} e^{-ix\xi} dx = \frac{\partial}{\partial t} \int_{-\infty}^{\infty} u(x, t) e^{-ix\xi} dx = \frac{\partial \hat{u}}{\partial t}(\xi, t)$$

where

$$\hat{u}(\xi, t) = \mathcal{F}_x u(\xi, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ix\xi} dx$$

Since the partial Fourier transform is done in x , the second term in Equation 1 can be rewritten with the Fourier transform rule for derivatives

$$\int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial x^2} e^{-ix\xi} dx = -\xi^2 \int_{-\infty}^{\infty} u(x, t) e^{-ix\xi} dx = -\xi^2 \hat{u}(\xi, t)$$

Putting this into the differential equation gives

$$\mathcal{F}_x \left(\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} \right) = \frac{\partial \hat{u}}{\partial t}(\xi, t) + \xi^2 \hat{u}(\xi, t) = 0$$

This has the general solution

$$\hat{u}(\xi, t) = c(\xi) e^{-\xi^2 t}$$

To determine $c(\xi)$, we set $t = 0$

$$\begin{aligned} c(\xi) = \hat{u}(\xi, 0) &= \int_{-\infty}^{\infty} u(x, 0) e^{-ix\xi} dx = \int_{-\infty}^{\infty} e^{-x^2} e^{-ix\xi} dx = \\ &= \mathcal{F} [e^{-x^2}] (\xi) = \sqrt{\pi} e^{-\frac{\xi^2}{4}} \end{aligned}$$

This gives

$$\hat{u}(\xi, t) = \sqrt{\pi} e^{(-\xi^2(t+\frac{1}{4}))} = \sqrt{\pi} e^{-\xi^2(t+\frac{1}{4})}$$

To get the solution we need to inverse transform this

$$\begin{aligned} \mathcal{F} [e^{-x^2}] &= \sqrt{\pi} e^{-\frac{\xi^2}{4}} \\ \mathcal{F} [e^{-(ax)^2}] &= \frac{1}{|a|} \sqrt{\pi} e^{-\frac{(\frac{\xi}{a})^2}{4}} \end{aligned}$$

From this we see we want $t + \frac{1}{4} = \frac{1}{4a^2} \rightarrow a = \sqrt{\frac{1}{4(t+\frac{1}{4})}} \rightarrow \frac{1}{a} = \sqrt{4t+1}$. With this, we get

$$\mathcal{F} \left[\frac{1}{\sqrt{4t+1}} e^{-\left(\frac{x}{\sqrt{4t+1}}\right)^2} \right] = \sqrt{\pi} e^{-\xi^2(t+\frac{1}{4})}$$

↓

$$u(x, t) = \frac{1}{\sqrt{4t+1}} e^{-\left(\frac{x}{\sqrt{4t+1}}\right)^2}$$

□

4.3

Solve the heat problem

$$\begin{aligned}
\frac{\partial u}{\partial t} - a \frac{\partial^2 u}{\partial x^2} &= -\alpha u, & t > 0, x \in \mathbb{R} \\
u(x, 0) &= \delta(x), & x \in \mathbb{R} \\
u(x, t) &\rightarrow 0, x \rightarrow \pm\infty, & t > 0
\end{aligned}$$

Solution:

This problem describes a heat problem where there is a sudden impulse of heat at $x = 0, t = 0$. Partial Fourier transforming gives

$$\begin{aligned}
&\int_{-\infty}^{\infty} \frac{\partial u}{\partial t} e^{-ix\xi} dx - a \int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial x^2} e^{-ix\xi} dx + \alpha \int_{-\infty}^{\infty} u e^{-ix\xi} dx \\
&\quad \downarrow \\
&\frac{\partial \hat{u}}{\partial t}(\xi, t) + a\xi^2 \hat{u}(\xi, t) + \alpha \hat{u}(\xi, t) = 0 \\
&\quad \downarrow \\
&\frac{\partial \hat{u}}{\partial t}(\xi, t) + (a\xi^2 + \alpha) \hat{u}(\xi, t) = 0 \\
&\quad \downarrow \\
&\hat{u}(\xi, t) = c(\xi) e^{-(a\xi^2 + \alpha)t}
\end{aligned}$$

To determine $c(\xi)$, set $t = 0$ in $\hat{u}(\xi, t)$

$$\hat{u}(\xi, 0) = \int_{-\infty}^{\infty} u(x, 0) e^{-ix\xi} dx = \int_{-\infty}^{\infty} \delta(x) e^{-ix\xi} dx = 1$$

This means

$$\hat{u}(\xi, t) = e^{-(a\xi^2 + \alpha)t} = e^{-a\xi^2 t} e^{-\alpha t}$$

Now $u(x, t)$ is given by the inverse transform of this

$$\begin{aligned}
\mathcal{F} [e^{-x^2}] &= \sqrt{\pi} e^{-\frac{\xi^2}{4}} \\
\mathcal{F} \left[\frac{e^{-\alpha t}}{\sqrt{\pi}} e^{-x^2} \right] &= e^{-\frac{\xi^2}{4}} e^{-\alpha t} \\
\mathcal{F} \left[\frac{e^{-\alpha t}}{2\sqrt{\pi at}} e^{-\left(\frac{x}{2\sqrt{a}}\right)^2} \right] &= e^{-\frac{(\xi 2\sqrt{at})^2}{4}} e^{-\alpha t}
\end{aligned}$$

Final answer

$$u(x, t) = \frac{e^{-\alpha t}}{\sqrt{4\pi at}} e^{-\frac{x^2}{4at}}$$

□

4.5

Find a solution to the Schrödingerequation

$$\frac{\partial \psi}{\partial t} - i \frac{\partial^2 \psi}{\partial x^2} = 0, \quad t > 0, x \in \mathbb{R}$$

that satisfy the initial value

$$\psi(x, 0) = \sin^2(x), \quad x \in \mathbb{R}$$

Solution:

We will partial Fourier transform the differential equation, which results in

$$\int_{-\infty}^{\infty} \frac{\partial \psi}{\partial t} e^{-ix\xi} dx + i\xi^2 \int_{-\infty}^{\infty} \psi e^{-ix\xi} dx = 0$$

$$\downarrow$$

$$\frac{\partial \hat{\psi}}{\partial t}(\xi, t) + i\xi^2 \hat{\psi}(\xi, t) = 0$$

$$\downarrow$$

$$\hat{\psi}(\xi, t) = c(\xi) e^{-i\xi^2 t}$$

To determine $c(\xi)$, we set $t = 0$ and get

$$\begin{aligned} \hat{\psi}(\xi, 0) = c(\xi) &= \int_{-\infty}^{\infty} u(x, 0) e^{-ix\xi} dx = \int_{-\infty}^{\infty} \sin^2(x) e^{-ix\xi} dx = \\ &= \int_{-\infty}^{\infty} \left(\frac{e^{ix} - e^{-ix}}{2i} \right)^2 e^{-ix\xi} dx = -\frac{1}{4} \int_{-\infty}^{\infty} (e^{2ix} - 2 + e^{-2ix}) e^{-ix\xi} dx \end{aligned}$$

$$\downarrow$$

$$c(\xi) = -\frac{1}{4} \mathcal{F} [e^{2ix} - 2 + e^{-2ix}] = -\frac{1}{4} [2\pi\delta(\xi - 2) - 4\pi\delta(\xi) + 2\pi\delta(\xi + 2)] =$$

$$\rightarrow c(\xi) = \frac{\pi}{2} [2\delta(\xi) - \delta(\xi - 2) - \delta(\xi + 2)]$$

This gives our $\hat{\psi}(\xi, t)$

$$\begin{aligned} \hat{\psi}(\xi, t) &= \frac{\pi}{2} [2\delta(\xi) - \delta(\xi - 2) - \delta(\xi + 2)] e^{-i\xi^2 t} = \\ &= \pi\delta(\xi) - \frac{\pi}{2} e^{-4it} \delta(\xi - 2) - \frac{\pi}{2} e^{-4it} \delta(\xi + 2) \end{aligned}$$

and our solution $\psi(x, t)$ is the inverse Fourier transform of $\hat{\psi}(\xi, t)$

$$\mathcal{F} \left[\frac{1}{2} - \frac{1}{4} e^{-4it} e^{2ix} - \frac{1}{4} e^{-4it} e^{-2ix} \right] = \pi\delta(\xi) - \frac{\pi}{2} e^{-4it} \delta(\xi - 2) - \frac{\pi}{2} e^{-4it} \delta(\xi + 2)$$

So our solution that satisfy the initial conditions is

$$\psi(x, t) = \frac{1}{2} - \frac{1}{4}e^{-4it}e^{2ix} - \frac{1}{4}e^{-4it}e^{-2ix} = \frac{1}{2} \left(1 - e^{-4it} \left[\frac{e^{2ix} + e^{-2ix}}{2} \right] \right)$$

↓

$$\psi(x, t) = \frac{1}{2}(1 - e^{-4it} \cos(2x))$$

□

4.6

Solve the problem

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} &= 0, & t > 0, x \in \mathbb{R} \\ u(x, 0) &= 0, \frac{\partial u}{\partial t}(x, 0) = \delta(x), & x \in \mathbb{R} \end{aligned}$$

Solution:

Partial Fourier Transform:

Partial Fourier transform gives

$$\frac{\partial^2 \hat{u}}{\partial t^2}(\xi, t) + \xi^2 \hat{u}(\xi, t) = 0$$

$$\hat{u}(\xi, t) = A(\xi) \cos(\xi t) + B(\xi) \sin(\xi t)$$

To determine $A(\xi)$ and $B(\xi)$, we set $t = 0$.

$$\hat{u}(\xi, 0) = \int_{-\infty}^{\infty} u(x, 0) e^{-ix\xi} dx = 0 \rightarrow A(\xi) = 0$$

$$\frac{\partial \hat{u}}{\partial t}(\xi, 0) = \int_{-\infty}^{\infty} \frac{\partial u}{\partial t}(x, 0) e^{-ix\xi} dx = \int_{-\infty}^{\infty} \delta(x) e^{-ix\xi} dx = 1$$

$$\frac{\partial \hat{u}}{\partial t}(\xi, 0) = \xi B(\xi) \cos(\xi \cdot 0) = \xi B(\xi) = 1 \rightarrow B(\xi) = \frac{1}{\xi}$$

We get our $\hat{u}(\xi, t)$ as

$$\hat{u}(\xi, t) = \frac{\sin(\xi t)}{\xi}$$

which has the inverse Fourier transform

$$\mathcal{F}^{-1} \left[\frac{\sin(\xi t)}{\xi} \right] = \frac{1}{2}(\theta(x+t) - \theta(x-t))$$

Partial Laplace Transform:

We can also solve it using the partial Laplace transform, which might be more convenient for wave equations. We set

$$U = \int_{-\infty}^{\infty} u(x, t) e^{-sx} dx$$

as the Laplace transform. Laplace transforming the differential equation in x gives

$$\int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial t^2} e^{-sx} dx - \int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial x^2} e^{-sx} dx = 0$$

$$\downarrow$$

$$\frac{\partial^2 U}{\partial t^2} - s^2 U = 0$$

$$\downarrow$$

$$U(s, t) = A(s)e^{st} + B(s)e^{-st}$$

To determine the constants $A(s)$ and $B(s)$, we set $t = 0$

$$U(s, 0) = A(s) + B(s) = \int_{-\infty}^{\infty} u(x, 0) e^{-sx} dx = 0 \rightarrow A(s) = -B(s)$$

$$\frac{\partial U}{\partial t}(s, 0) = sA(s) - sB(s) = \int_{-\infty}^{\infty} \frac{\partial u}{\partial t}(x, 0) e^{-sx} dx = \int_{-\infty}^{\infty} \delta(x) e^{-sx} dx = 1$$

$$\downarrow$$

$$sA(s) - sB(s) = 1 \rightarrow 2sA(s) = 1 \rightarrow A(s) = \frac{1}{2s}, B(s) = -\frac{1}{2s}$$

$$\downarrow$$

$$U(s, t) = \frac{1}{2s}(e^{st} - e^{-st})$$

Now all that's left to do is find the inverse Laplace transform of $U(s, t)$:

$$\mathcal{L} [\theta(x)] = \frac{1}{s}$$

$$\mathcal{L} [\theta(x-a)] = \frac{1}{s} e^{-as}$$

$$\mathcal{L} \left[\frac{1}{2}(\theta(x+t) - \theta(x-t)) \right] = \frac{1}{2s}(e^{st} - e^{-st})$$

$$\downarrow$$

$$u(x, t) = \frac{1}{2}[\theta(x+t) - \theta(x-t)]$$

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Solution:

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